CrunchBite Digital — Technical Context

Platform Architecture

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│                              CLIENT LAYER                                 │
│                                                                           │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  │
│  │  Mobile App  │  │  Web Order   │  │    Kiosk     │  │  Franchise /  │  │
│  │ (React Native│  │  (Next.js 14 │  │  (Electron + │  │  Store Manager │  │
│  │  iOS/Android)│  │   SSR / RSC) │  │   React)     │  │  Dashboard    │  │
│  └──────┬───────┘  └──────┬───────┘  └──────┬───────┘  │  (React SPA)  │  │
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          │  HTTPS + WSS    │                 │  local TCP/WS  │ HTTPS
┌─────────▼─────────────────▼─────────────────▼────────────────▼───────────┐
│                  API GATEWAY — AWS API Gateway + CloudFront CDN           │
│        JWT (Cognito)  •  Rate limits  •  WAF OWASP rules  •  mTLS (B2B)  │
└──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──┘
       │      │      │      │      │      │      │      │      │      │
┌──────▼──┐ ┌─▼──────┐ ┌─▼──────┐ ┌─▼──────┐ ┌─▼──────┐
│ Identity │ │ Order  │ │ Menu   │ │Loyalty │ │Notification│
│ Service  │ │Service │ │Service │ │Service │ │Service     │
│          │ │        │ │        │ │        │ │            │
│Cognito + │ │PostgreSQL│ │DynamoDB│ │PostgreSQL│ │Braze SDK + │
│Lambda    │ │+ Lambda│ │+ Lambda│ │+ Lambda│ │SQS + Lambda│
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       │      │      │      │      │      │      │      │      │
┌──────▼──────▼──────▼──────▼──────▼──────▼──────▼──────▼──────▼──────┐
│                  DOMAIN EVENT BUS — Apache Kafka (MSK)                 │
│  order.placed  •  order.ready  •  item.86d  •  points.earned  •  offer.used │
│  payment.completed  •  delivery.dispatched  •  shift.opened  •  menu.updated│
└──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──┘
       │      │      │      │      │      │      │      │      │      │
┌──────▼──┐ ┌─▼──────┐ ┌─▼──────┐ ┌─▼──────┐ ┌─▼──────┐
│ Kitchen  │ │Payment │ │Payment │ │Payment │ │Payment │
│ Service  │ │Service │ │Service │ │Service │ │Service │
│          │ │        │ │        │ │        │ │        │
│KDS app on│ │Adyen API│ │Adyen API│ │Adyen API│ │Adyen API│
│Epson HW  │ │+ Lambda│ │+ Lambda│ │+ Lambda│ │+ Lambda│
│WebSocket │ │        │ │        │ │        │ │        │
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Services Reference

Service	Responsibility	Technology
Identity Service	Customer registration, login, MFA, guest checkout tokens, social sign-in	AWS Cognito + Lambda (Node.js 20)
Order Service	Cart management, order lifecycle (placed → in-kitchen → ready → collected), POS sync	Lambda + PostgreSQL 15 (Aurora Serverless v2)
Menu Service	Menu catalogue, daypart scheduling, LTO activation, 86-ing propagation, allergen data	Lambda + DynamoDB + ElastiCache Redis
Loyalty Service	Points earn/redeem, tier calculation, offer validation, fraud velocity checks	Lambda + PostgreSQL 15
Payment Service	Adyen session creation, capture, refund, saved card token management	Lambda + Adyen Terminal API
Kitchen Service	KDS ticket delivery, station routing, bump/recall, SOS timer	Lambda + WebSocket API Gateway + Epson KDS
Notification Service	Push notifications (FCM/APNs), SMS (Twilio), email (SendGrid), in-app inbox	Lambda + SQS FIFO + Braze SDK
Analytics Service	Real-time SOS metrics, item-level sales, franchise roll-up, Tableau feed	Kafka → Snowflake Connector
Delivery Gateway	Otter webhook ingestion, order injection, ETA proxy	Lambda + Otter REST API
Store Config Service	Store hours, delivery radius, price overrides, feature flags per store	Lambda + DynamoDB
Catering Service	Group order management, coordinator assignment, lead-time validation	Lambda + PostgreSQL 15
Franchise Portal BFF	Backend-for-frontend for franchise dashboard; multi-store aggregation	Lambda + GraphQL (Apollo Server)

Tech Stack

Customer-Facing Frontend

- **Mobile**: React Native 0.74 (iOS 16+, Android 10+); Expo managed workflow
- **Web ordering**: Next.js 14 (App Router, React Server Components); deployed to CloudFront + S3
- **Kiosk**: Electron 30 wrapping a React app; runs on Windows 10 IoT; offline-capable (SQLite local cache for menu)
- **Franchise / Manager Dashboard**: React 18 SPA + Apollo Client; Tailwind CSS + Radix UI components
- **State management**: Zustand (local UI state); React Query v5 (server state); Jotai (kiosk-specific atom store)

Backend

- **Runtime**: Node.js 20 LTS on AWS Lambda (arm64 Graviton3)
- **API style**: REST (customer-facing); GraphQL (franchise BFF); WebSocket (KDS real-time)

- **Auth**: JWT RS256 via Cognito User Pools; anonymous guest token (scoped JWT, 4-hour TTL) for unauthenticated ordering
- **Primary database**: Amazon Aurora PostgreSQL 15 Serverless v2 (Order, Loyalty, Catering services)
- **Document / catalogue store**: DynamoDB (Menu Service — low-latency read, global tables for multi-region)
- **Cache**: ElastiCache Redis 7 — menu cache (TTL 60 s), session store, idempotency keys for payment
- **Event streaming**: Amazon MSK (Kafka 3.6) — domain events, SOS metrics, Snowflake CDC
- **Queue**: SQS FIFO (Notification Service dispatch, order injection from Otter)

COTS Products in the Stack

Product	Role	Integration point
Oracle MICROS POS	In-store POS, receipt printing, drawer management	Order Service → MICROS REST API + TCP socket
Adyen	All payment processing (app, web, kiosk, drive-through)	Payment Service → Adyen REST + Terminal API
Braze	Campaign management, segmented push/email/SMS, A/B testing	Notification Service + Loyalty Service → Braze REST SDK
Otter	Delivery order aggregation (Uber Eats, DoorDash, Grubhub)	Delivery Gateway ← Otter webhook
Infor CloudSuite F&B	Inventory, procurement, recipe/PIM	Menu Service ← Infor REST API (allergen, nutrition)
Revel Systems	Kiosk software layer; wraps MICROS	Kiosk App → Revel API
Epson KDS	Kitchen Display hardware	Kitchen Service → WebSocket push
Medallia	NPS and guest CSAT surveys triggered post-transaction	Analytics Service → Medallia REST
Snowflake	Enterprise data warehouse; franchise and ops analytics	Kafka → Snowflake Kafka Connector
Tableau	Franchise dashboards, SOS trend analysis	Snowflake → Tableau

Infrastructure & DevOps

- **Cloud**: AWS (us-east-1 primary; us-west-2 failover for Order + Payment services)
- **IaC**: AWS CDK v2 TypeScript; all resources version-controlled; no click-ops
- **CI/CD**: GitHub Actions — lint → test (unit + integration) → E2E (Playwright) → deploy staging → manual promote to prod
- **Container registry**: Amazon ECR; Lambda deployed as container images
- **Secrets**: AWS Secrets Manager; Adyen API key, Braze REST key, MICROS credentials rotated quarterly
- **Observability**: AWS CloudWatch + X-Ray; Datadog APM for cross-service tracing; PagerDuty for P1/P2 alerts
- **Feature flags**: LaunchDarkly — used to gate LTO activations, new menu features, and kiosk UI experiments

Key Architectural Constraints

- **86-ing propagation SLA**: item unavailability must reach all channels (app, web, kiosk, MICROS) within **60 seconds** via Kafka → Menu Service → cache invalidation
- **Kiosk offline mode**: kiosk must serve menu and accept orders during 5-minute network blip; local SQLite cache; orders sync when connectivity restored
- **Idempotency**: all payment and points-earning operations keyed by `orderId`; duplicate Kafka events or retried API calls must not double-charge or double-award
- **MICROS integration**: MICROS POS is the authoritative receipt and tax engine; Order Service always confirms with MICROS before marking order `COMPLETE`
- **Otter webhook ingestion**: Otter sends delivery orders as webhooks; Delivery Gateway must acknowledge within **3 seconds** or Otter retries; idempotency key = Otter order ID

Non-Functional Requirements

Category	Requirement
Availability	99.95% for Order, Payment, Menu services during peak trading hours (11 AM – 2 PM, 5 PM – 8 PM local)
Latency	p95 order placement < 800 ms end-to-end; menu load < 300 ms (CDN-cached)
Throughput	2.4 M transactions/day peak; burst capacity for limited-time promotions (3× normal)
Kiosk offline	Accept orders for up to 5 minutes without network; sync on reconnect
86-ing propagation	≤ 60 seconds from manager action to all channel updates
Data retention	Transaction records: 7 years; loyalty point history: 3 years after account closure
Test coverage	Unit ≥ 80%; integration ≥ 60%; Playwright E2E on critical ordering paths